1) During training, the trainingsdata is separated into several **batches**, which are nothing else than subsets of the whole trainingsdata. These batches are used for a single training step (evaluating the gradient and applying the gradient).
If all batches are processed once, an **epoch** of the training process passed.
All available data are separated into three parts: trainingsdata, validationdata and testdata.
During each epoch, the **trainingsdata** is used for evaluating the gradient and applying the learning algorithm.
After each epoch, the **validationdata** is used for checking, if the learning process is heading to the right direction. By using separate dataset for evaluating the gradient (trainingsdata) and for checking the progress of the learning process (validationdat), the danger of overfitting is minimized.
After the learning process, the finalized Neural Network is evaluated by the **testdata**. If the results on the testdata are promising, the training process can be assumed to be reliable.
In the case of **overfitting**, the Neural Network performs very well on the trainingsdata and maybe on the validationdata but not very well on the testdata. Overfitting in simple words means: the Neural Network has learned the trainingsdata very well and is too highly specialised to these data and has not learned general rules of classification.

2) Simplest possible classificator always chooses the class with highest probability: bears

$\Rightarrow$ correct : 987
incorrect: 356 + 1

$$\Rightarrow \text{accuracy} = \frac{987}{987 + 356 + 1} = 0{,}734 = 73{,}4\,\%$$

3)

$$CM = \begin{pmatrix} 8 & 1 & 2 \\ 0 & 5 & 3 \\ 3 & 2 & 5 \end{pmatrix}$$

$$\text{accuracy} = \frac{\text{Sum of main diagonal}}{\text{Sum of all elements}} = \frac{8+5+5}{8+1+2+5+3+3+2+5}$$

$$= \frac{18}{29} = 62{,}1\,\%$$

$$\text{precision}_0 = \frac{8}{8+3} = \frac{8}{11}$$

$$\text{precision}_1 = \frac{5}{5+1+2} = \frac{5}{8}$$

$$\text{precision}_2 = \frac{5}{5+2+3} = \frac{1}{2}$$

$$\text{recall}_0 = \frac{8}{8+1+2} = \frac{8}{11}$$

$$\text{recall}_1 = \frac{5}{5+3} = \frac{5}{8}$$

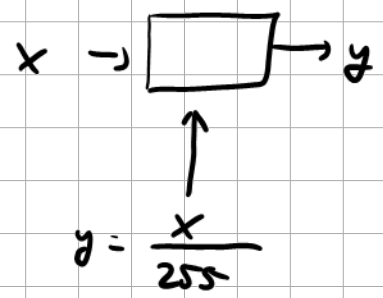
$$\text{recall}_2 = \frac{5}{5+2+3} = \frac{1}{2}$$

SpokenWordIndex is used as row-index in the confusion matrix.
The simplest possible classificator uses the index, with the highest row-sum, which is the 0-th row.
The accuracy for this case is:

$$\frac{11}{11+8+10} = \frac{11}{29} = 0.379 \approx \frac{1}{3}$$

$\Rightarrow$ this dataset can be assumed to be balanced.

4) Uniform distribution



Normal distribution: zero mean and variance = 1

$$\tilde{y}_{(n)} = x_{(n)} - a$$

$$\sum_{n=0}^{N-1} \tilde{y}_{(n)} \overset{!}{=} 0 \quad \Leftrightarrow \quad \sum_{n=0}^{N-1} (x_{(n)} - a) = 0 \quad \Leftrightarrow \quad \sum_{n=0}^{N-1} x_{(n)} = N \cdot a$$

$$\Leftrightarrow \quad a = \frac{1}{N} \sum_{n=0}^{N-1} x_{(n)}$$

$\Rightarrow$ $\tilde{y}_{(n)}$ has zero mean

$y_{(n)} = b \cdot \tilde{y}_{(n)}$ multiplication by b: mean value remains zero

$$\frac{1}{N} \sum_{n=0}^{N-1} y_{(n)}^2 \overset{!}{=} 1 \quad (\text{variance} \overset{!}{=} 1)$$

$$\Leftrightarrow \quad \sum_{n=0}^{N-1} b^2 \tilde{y}_{(n)}^2 = N \quad \Leftrightarrow \quad b^2 \sum_{n=0}^{N-1} \tilde{y}_{(n)}^2 = N$$

$$\Leftrightarrow \quad b = \sqrt{\frac{N}{\sum_{n=0}^{N-1} \tilde{y}_{(n)}^2}}$$

$\Rightarrow$ $y_{(n)}$ has zero mean and unit variance.